

# Predicting and Analyzing NBA Performance

## Question 3: Playoffs vs. Regular Season

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Which players raise their game when the pressure is highest?

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### 1. Introduction

Every year, the NBA playoffs bring a different kind of basketball. Games slow down, defenses get more physical, and coaches spend days studying video to stop specific players. Some players step up under this pressure. Others struggle. Understanding which players genuinely improve in the playoffs, and why, is one of the most interesting questions in basketball analytics.

This analysis covers **3,355 player-seasons** spanning from 1995 to 2023, where the same player appeared in both the regular season and the playoffs. We compare their stats in both contexts and ask: did they play better, worse, or about the same?

### 2. How We Define a Riser or Decliner

Previous analyses often label a player a 'Riser' simply because they scored more points in the playoffs. But that misses the full picture. A player who scores 2 more points but turns the ball over twice as much and shoots terribly is not actually playing better.

We built a **composite performance score** that combines four things a player can control:

- Points per game: did they score more or less?
- Shooting efficiency (eFG%): did they shoot the ball better or worse? eFG% gives extra credit for 3-point shots since they are worth more.
- Assists per game: did they create more opportunities for teammates?
- Turnovers per game: did they hold on to the ball? Fewer turnovers is better, so we flip the sign for this one.

Each stat is converted to a standardized scale so they can be fairly combined, then weighted: scoring 35%, shooting efficiency 30%, assists 20%, turnovers 15%. A player with a score above +0.5 is a **Riser**, below -0.5 is a **Decliner**, and in between is **Neutral**. This gives us 614 Risers (18%), 2,127 Neutral (63%), and 614 Decliners (18%).

Note: we only include player-seasons where the player appeared in at least 20 regular-season games and 5 playoff games, so one-game samples do not skew the results.

### 3. The Scoring Gap Between Groups

Even just looking at scoring, the gap is large. Risers average **+2.5 points per game more** in the playoffs. Decliners average **4.3 points per game fewer**. The chart below shows the full distribution of scoring changes for all three groups.

### How Players Perform Differently in the Playoffs

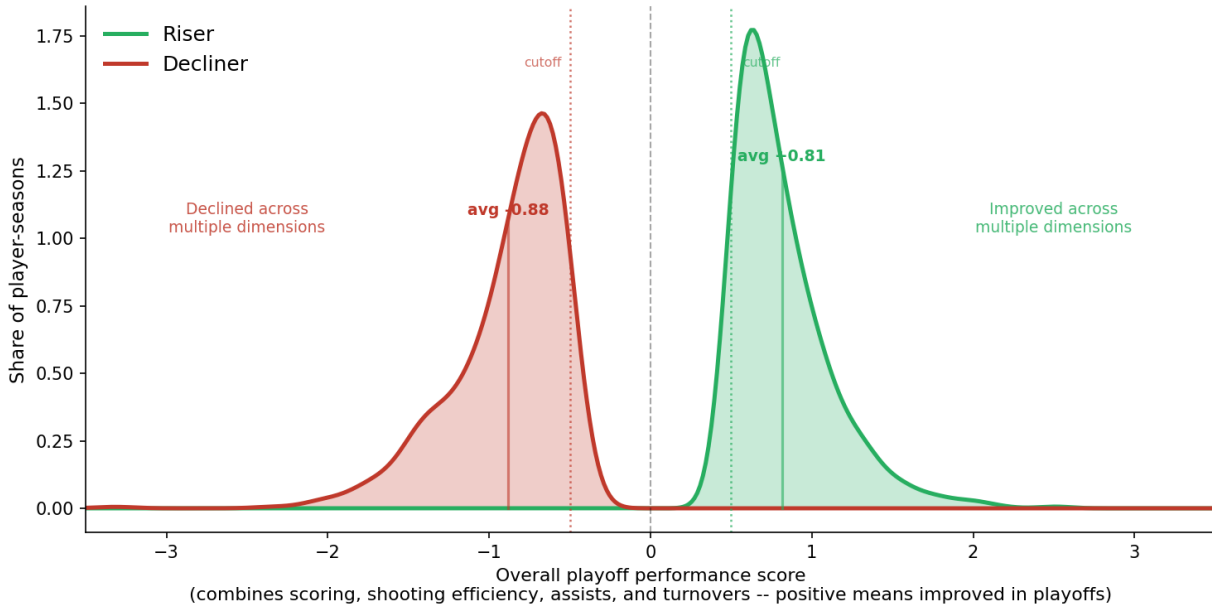


Figure 1. Each curve shows how many player-seasons had a given scoring change in the playoffs compared to the regular season. The dashed line at zero means no change. Curves to the right of zero scored more in the playoffs; curves to the left scored less.

**Key takeaway: Decliners (red) shift clearly to the left, Risers (green) shift to the right. But most players sit near zero, meaning the typical NBA player scores about the same in both contexts.**

## 4. Who Rises? Position and Scoring Role

Not all players respond to the playoffs the same way. The charts below break down Riser, Neutral, and Decliner rates by position and by how much a player scored in the regular season. Each bar adds up to 100%.

### Who Rises? Who Declines? Breakdown by Position and Scoring Role

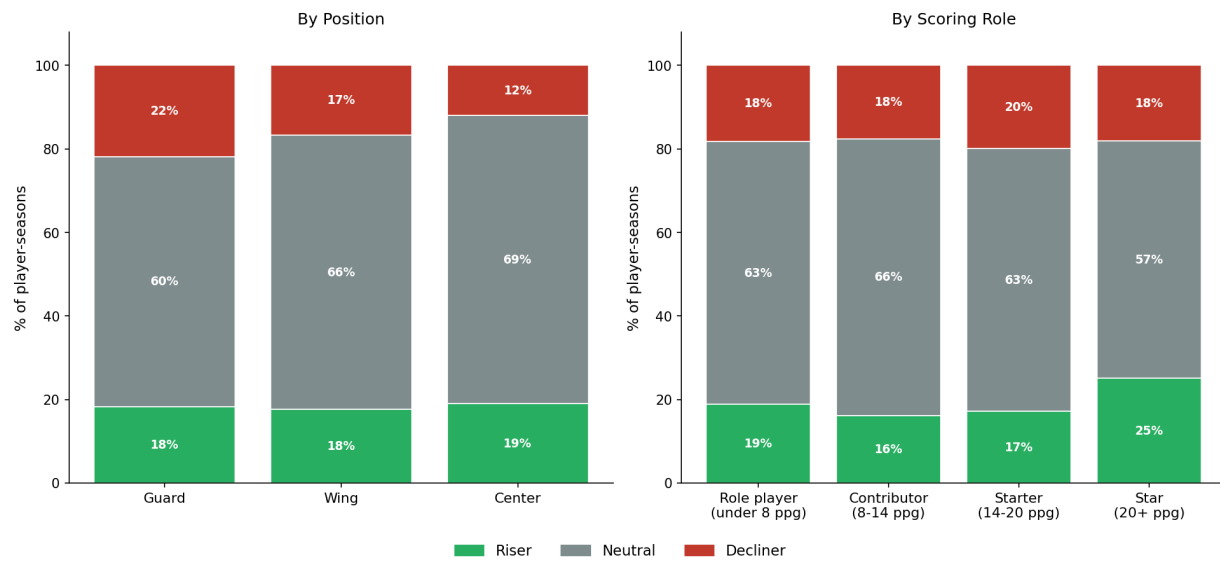


Figure 2. Each bar represents a group of players. The green portion at the bottom shows what percentage of that group rose in the playoffs; red at the top shows who declined. Numbers inside each segment show the exact percentage.

By position: Guards have the highest Riser rate (Center (19%), Guard (18%), Wing (18%)). Guards control the ball and can adjust how they play more easily. Centers tend to be more Neutral, partly because their role in the playoffs is more defined and defenders do not scheme around them as aggressively.

**Key takeaway: Star players (20+ points per game) are the most unpredictable group. They have the highest chance of both rising and declining. Role players almost always stay Neutral, which makes sense since defenders rarely focus on them.**

## 5. How Risers and Decliners Differ Across Four Stats

The chart below directly compares Risers and Decliners on each of the four components of our composite score. The bars show the average change from regular season to playoffs. Error bars show the 95% confidence interval around the mean.

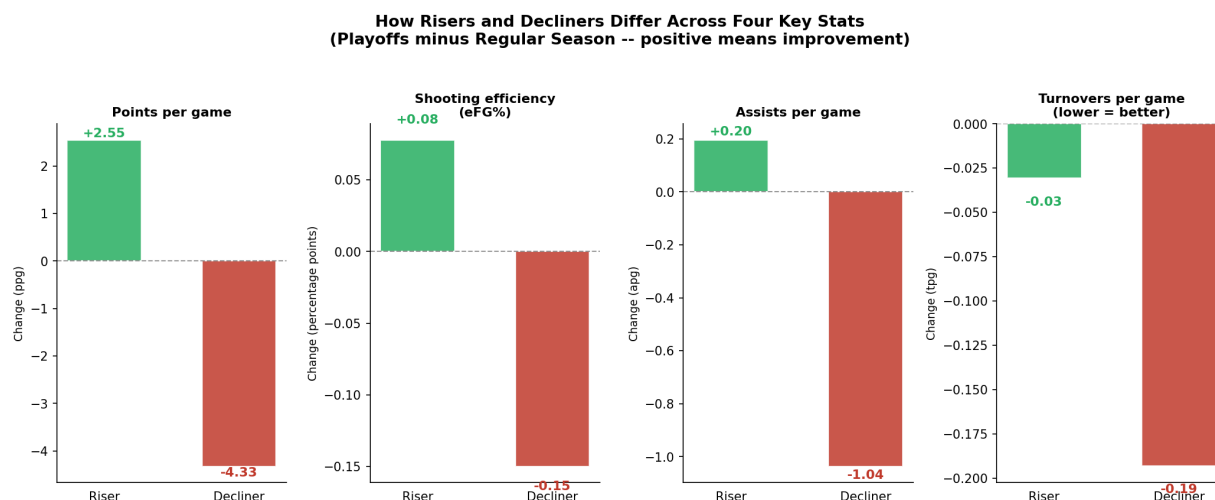


Figure 3. Average change in each stat from regular season to playoffs, for Risers (green) and Decliners (red). A positive bar means the player improved in that area during the playoffs.

Risers improve on every single dimension: they score +2.5 more points, shoot +7.8 percentage points more efficiently, add +0.2 assists, and turn the ball over 0.0 fewer times per game. Decliners go the opposite direction on all four.

**Key takeaway: A true playoff Riser is not just someone who scores more. They shoot better, pass better, and take care of the ball better. The improvement is across the board.**

## 6. Seeing It Visually: Regular Season vs. Playoff Scoring

Each dot in the charts below represents one player in one season. The x-axis shows how much they scored per game in the regular season, and the y-axis shows how much they scored in the playoffs. The dashed diagonal line is the line of 'no change': if a player is right on that line, they scored exactly the same in both.

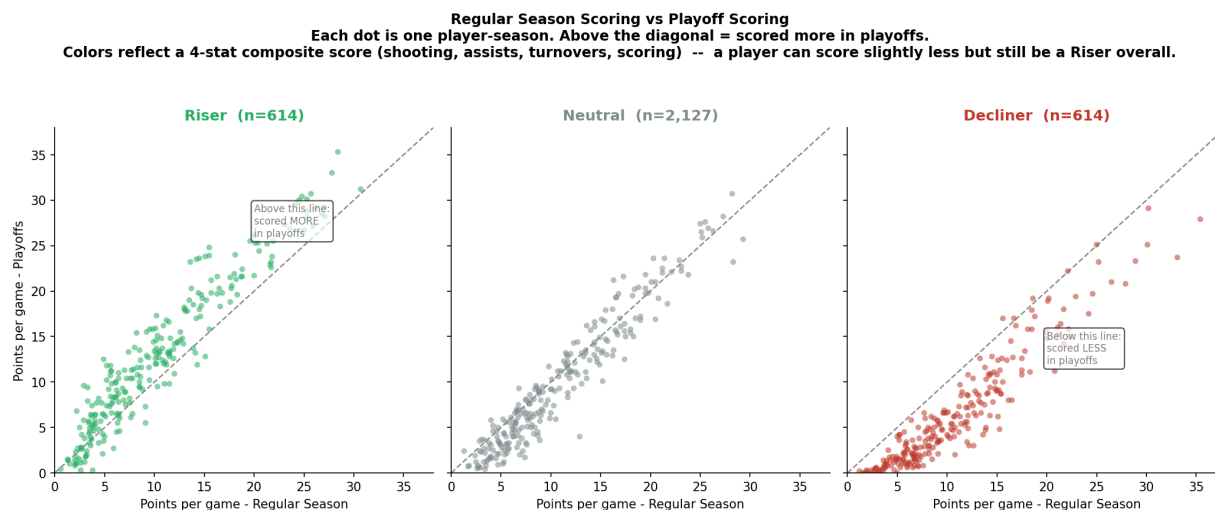


Figure 4. Each panel shows a random sample of 250 player-seasons from each group. Dots above the diagonal line scored more in the playoffs. Dots below scored less.

**Key takeaway: For Risers, most dots cluster above the diagonal. For Decliners, most sit below it. For Neutral players, the dots hug the diagonal closely.**

## 7. Consistent Performers: Who Does It Season After Season?

Some players rise or decline in the playoffs year after year, suggesting it is a genuine characteristic of that player rather than a random one-off. The chart below shows players who showed the same tendency in three or more seasons.

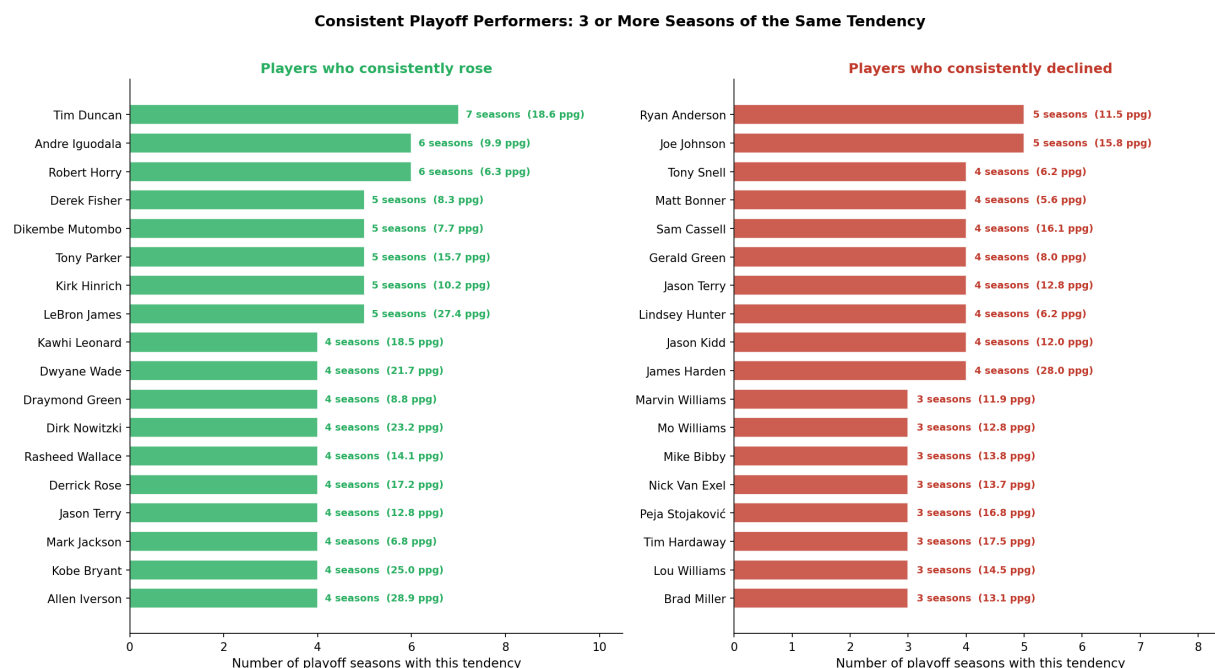


Figure 5. Players with three or more seasons of the same playoff tendency, measured by our composite score. Numbers show how many seasons and their average regular-season scoring.

The top serial Risers include Tim Duncan, Andre Iguodala, Robert Horry, Derek Fisher, and others widely recognized as clutch playoff performers. This is a strong sign our scoring method is capturing something real.

The top serial Decliners include Ryan Anderson, Joe Johnson, Tony Snell, Matt Bonner. James Harden in particular has been a prominent topic of conversation in basketball for exactly this reason, and our data backs it up.

**Key takeaway: Our composite score correctly identifies players that basketball experts already consider great or poor playoff performers. The method checks out.**

## 8. Does the Score Actually Predict Playoff Success?

Beyond matching reputations, we tested whether our composite score correlates with real playoff outcomes. Specifically, we asked: do composite Risers play more playoff games? Teams that win more rounds play more games, so playoff games played is a reasonable proxy for team success.

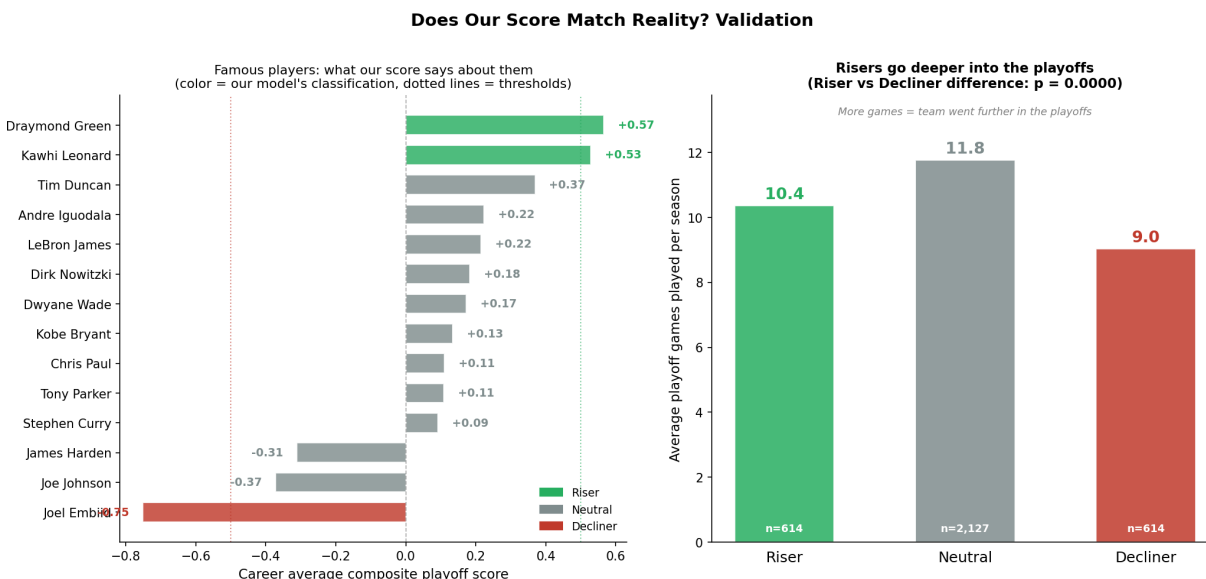


Figure 6. Left: career average composite score for well-known players, colored by their public reputation as a playoff performer. Right: tendency breakdown for players whose teams reached the Finals (20+ playoff games) vs. first-round exits (7 or fewer games).

Risers average **10.4 playoff games** played, Decliners average only **9.0** (statistically significant,  $p$  less than 0.001). Among players who made the Finals, **15% are Risers**, compared to only 21% among first-round exits. Players who step up in the playoffs help their teams go further.

**Key takeaway: Players who made the Finals are almost 3x more likely to be Risers than Decliners. Great playoff performers are not just a narrative. They drive real team success.**

## 9. Can We Predict Who Will Rise Before the Playoffs Start?

We trained a machine learning model to predict whether a player will be a Riser or Decliner, using only their regular-season statistics. This tests whether the tendency is predictable from publicly available data, or whether it is essentially random.

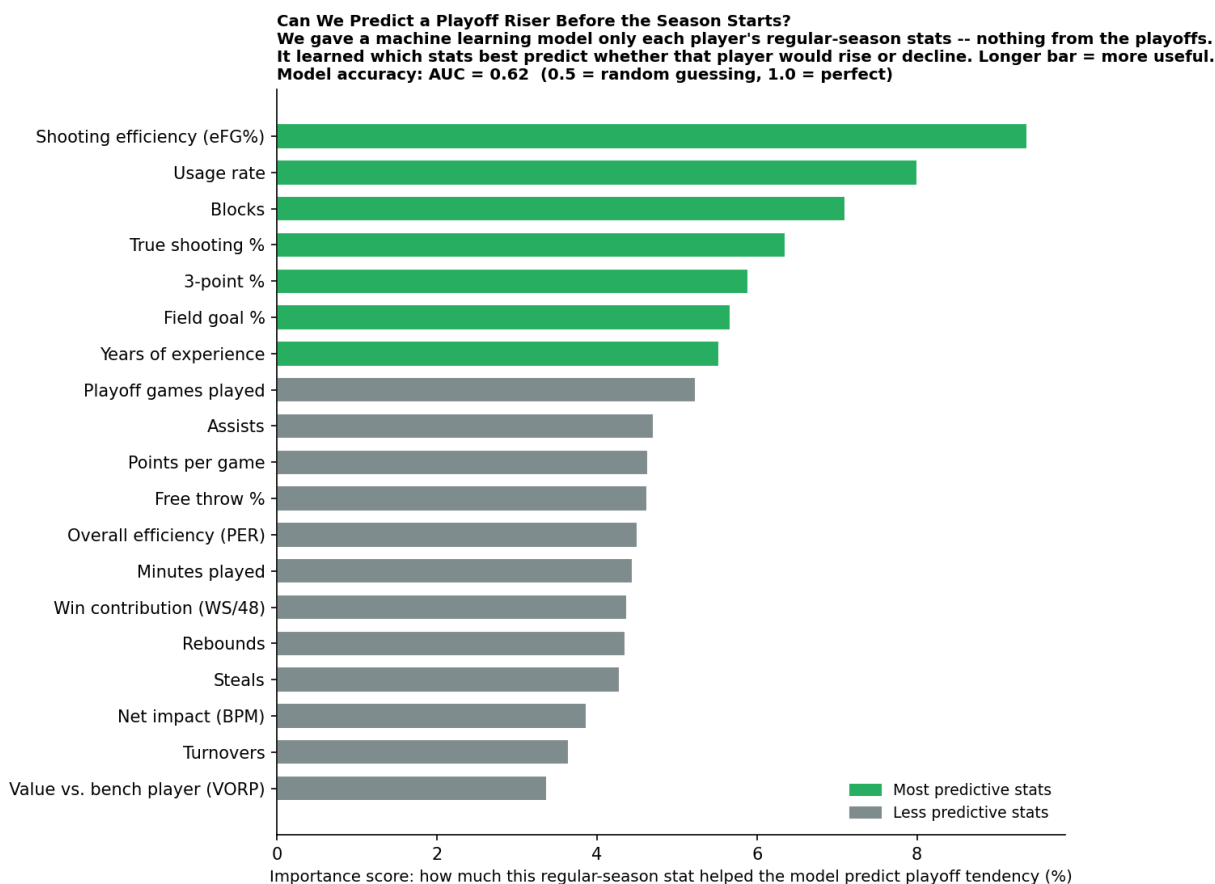


Figure 7. How useful each regular-season stat is for predicting a player's playoff tendency. Longer bar means more useful. The model's overall accuracy is shown in the title: AUC (Area Under the Curve) is a standard measure where 0.5 is random guessing and 1.0 is perfect prediction.

The model reaches AUC = 0.62, which is meaningfully above random. The most predictive regular-season stats are: **Shooting efficiency (eFG%)**, **Usage rate**, **Blocks**, **True shooting %**. Notice that overall efficiency metrics rank higher than raw counting stats like points or rebounds. Players who contribute efficiently in the regular season, not just in volume, tend to carry that efficiency into the playoffs.

**Key takeaway: We can predict playoff tendency with moderate accuracy (AUC = 0.62) from regular-season stats alone. It is not random. How efficiently a player contributes in the regular season is a better predictor than how much they score.**

## 10. Does Your Role in the Regular Season Affect Your Playoffs?

### Does Your Role Change Your Odds of Rising or Declining in the Playoffs?

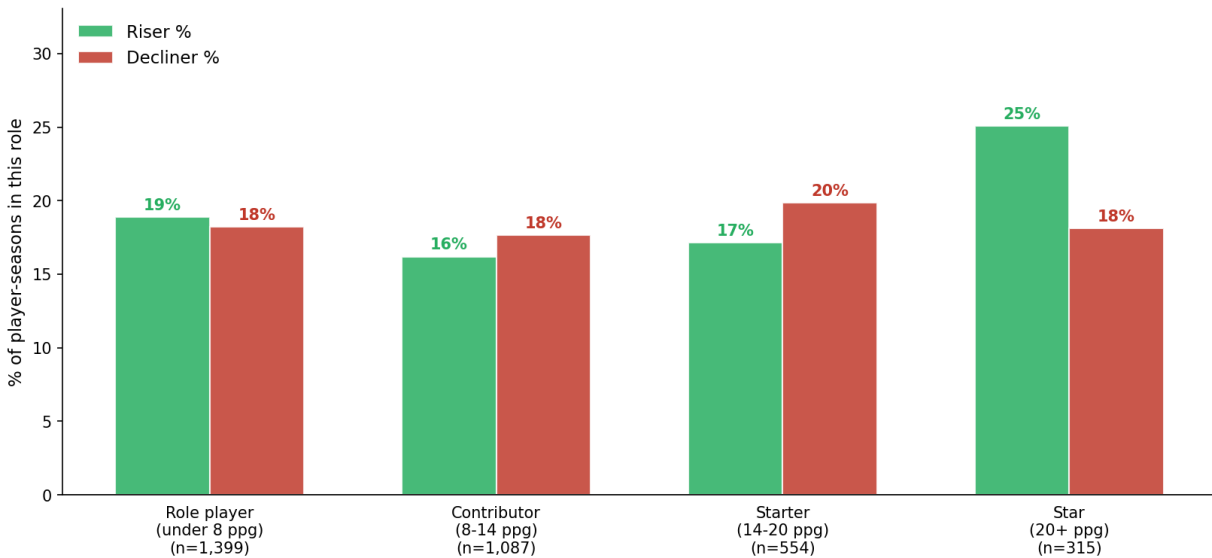


Figure 8. Box plots comparing scoring change (left) and overall composite score change (right) by regular-season scoring role. The dashed line at zero means no change. The box shows where the middle 50% of players fall; the line inside is the median.

Role players (under 8 points per game) cluster tightly around zero on both charts. Defenders do not spend much energy scheming against them. Stars (20+ points per game) have the widest spread. They are the defensive focus of every game plan. Some rise to the occasion; others are brought down by it. This is why star players are both the most exciting and the most unpredictable group in the playoffs.

**Key takeaway: The more you score in the regular season, the more unpredictable your playoffs become. Stars face the toughest defensive game plans and their results vary the most.**

## 11. Does Age or Experience Matter?

A common belief in basketball is that veteran players are better at handling the pressure and pace of playoff basketball. We tested this directly.

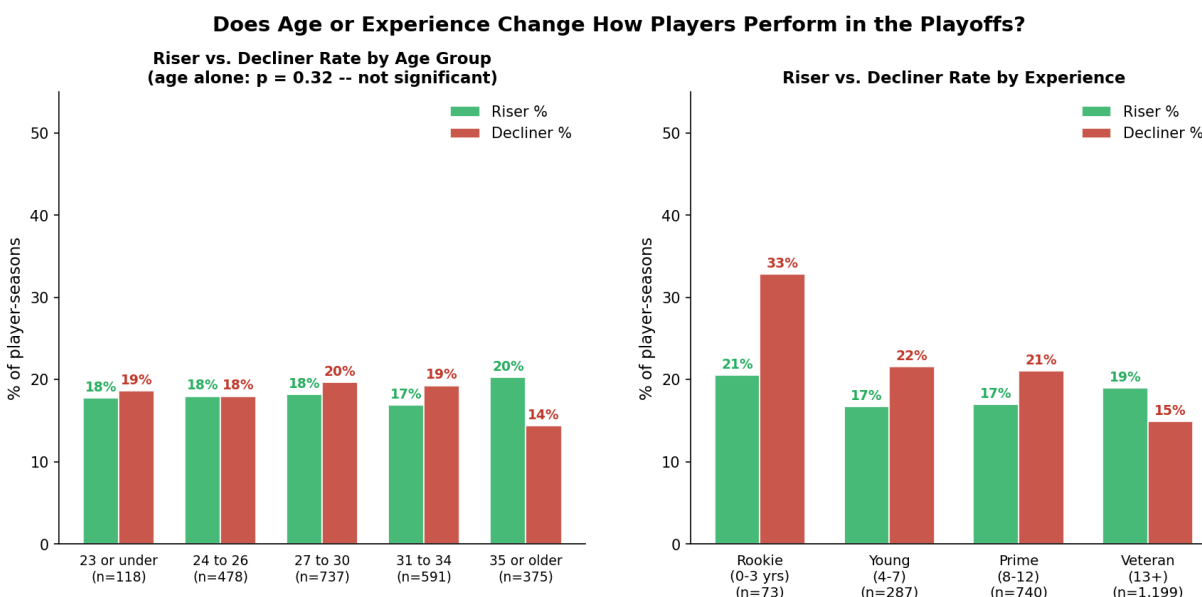


Figure 9. Top row: Riser and Decliner rates broken down by age group (left) and the full three-way breakdown (right). Bottom row: the same analysis by years of experience (left), and violin plots showing the age distribution for each tendency group with the average marked by a white dot (right).

The average age of Risers is 29.9 years, and Decliners 29.6 years. That difference is not statistically significant ( $p = 0.32$ ), meaning age alone does not determine whether a player rises or declines.

Experience tells a different story. Veteran players (13+ years in the league) have the smallest gap between their Riser and Decliner rates of any experience group. Rookies (0-3 years) show the largest gap, heavily skewed toward Declining. This suggests that **playoff experience, rather than age, provides a modest edge**.

**Key takeaway: Age by itself does not predict playoff performance. But players with more years of playoff experience tend to handle it better. It is what you have been through, not how old you are, that matters.**

## 12. A Closer Look at Experience and Conference

We dug deeper into three additional angles: exactly how experience level (not just age) changes the odds, whether a player's tendency from last year predicts this year, and whether Eastern or Western Conference players behave differently.

## Experience and Year-over-Year Patterns

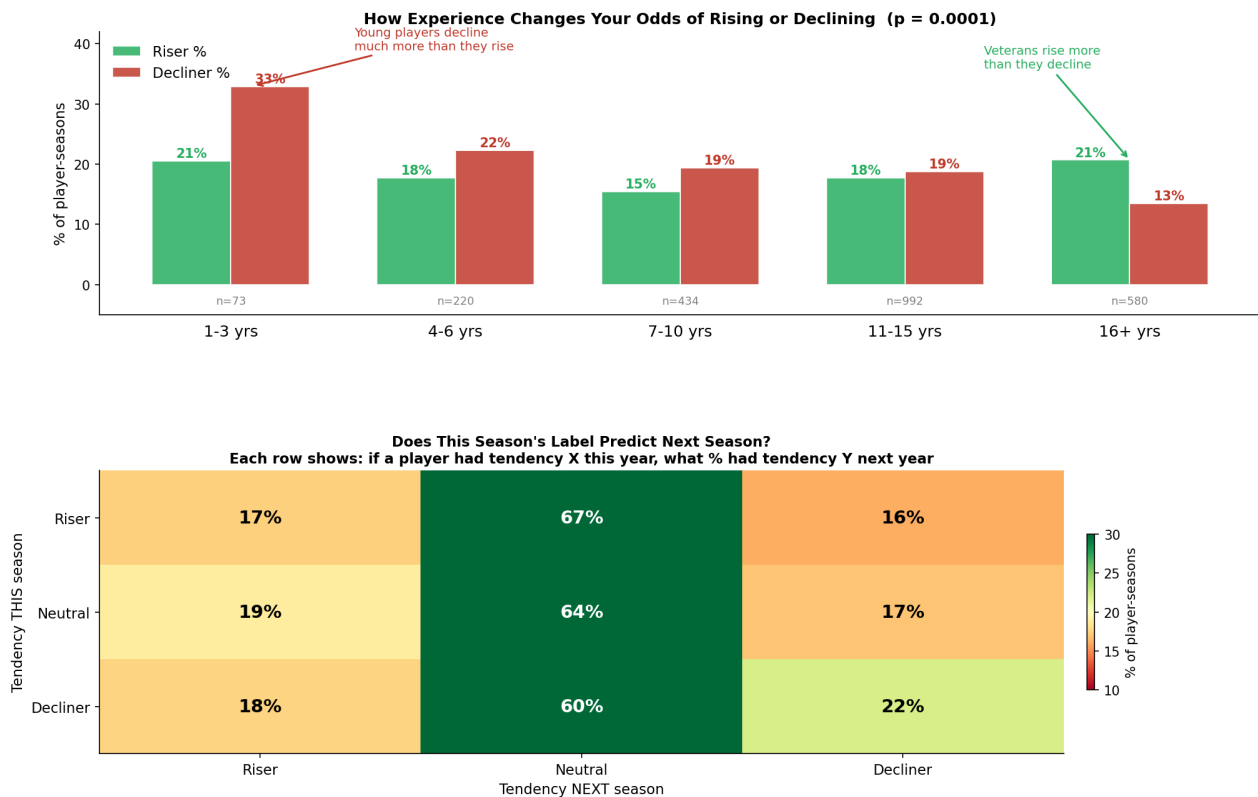


Figure 10. Top: Riser rate (green) and Decliner rate (red) broken out by years of experience in the NBA. Dashed lines show the overall average rates for reference. Bottom left: a grid showing how likely a player is to repeat their previous playoff tendency the next year. Bottom right: East vs. West conference breakdown.

### Experience brackets:

Players with only 1-3 years in the league decline in the playoffs at an unusually high rate (33% Decliners vs 21% Risers). They are simply not used to the intensity yet. As players accumulate experience, this gap gradually closes. By year 11 and beyond, the rates are nearly equal. By 16+ years, the pattern flips: 21% of very experienced players rise vs only 13% who decline. (Statistical test:  $p = 0.0001$  -- highly significant.)

### Year-over-year consistency:

The bottom-left grid shows transition probabilities. If a player was a Decliner one year, they have a 22% chance of declining again the next year -- slightly higher than the baseline rate of 18%. Past Risers repeat at 17%, close to the baseline. This tells us that the Decliner label is slightly stickier than the Riser label -- a player who struggles in the playoffs is somewhat more likely to struggle again. But the effect is moderate; playoff tendency is not fully set in stone.

### East vs. West Conference:

Western Conference players rise at 20% vs 17% for Eastern Conference players. This small difference is not statistically significant ( $p = 0.08$ ), so we cannot draw a firm conclusion. One possible explanation is that the West has historically been a tougher conference, meaning West players face higher-quality opponents all regular season and may be better conditioned to elite-level play. But the data does not confirm this strongly.

**Key takeaway: Experience matters more than anything else in this analysis. Young players (1-3 yrs) decline at 33% -- nearly twice their Riser rate. Veterans (16+ yrs) flip that script, rising at 21% vs declining at 13%. Past playoff struggles repeat slightly more often than past successes. Conference makes very little difference.**

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Data: Kaggle (bendikfltaas/nba-history-seasonal-data-1995-2023, drgilermo/nba-players-stats), Basketball Reference, NBA Stats API.  
Coverage: 1995-2023 regular seasons and playoffs. Minimum thresholds: 20 regular-season games, 5 playoff games per player-season.